機械力学 I 解答用紙

week - 11 減衰強制振動（速度の応答倍率，加速度の応答倍率）

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m11c04B0102

(1)	$\zeta = \frac{c}{2\sqrt{mk}} = 0.245$
(2)	$\omega_n = \sqrt{\frac{k}{m}} = 4.90$ [rad/s]
(3)	$r = \frac{\omega}{\omega_n} = 1.22$
(4)	$M_v = \frac{r}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} = 1.58$

m11c04B0202

(1)	$\zeta = \frac{c}{2\sqrt{mk}} = 0.245$
(2)	$\omega_n = \sqrt{\frac{k}{m}} = 4.90$ [rad/s]
(3)	$M_{vp} = \frac{1}{2\cdot\zeta} = 0.490$
(4)	$\omega = \omega_n = 4.90$ [rad/s]

m11c04B0301

(1)	$\zeta = \frac{1}{2M_{vp}} = 0.167$
(2)	$c = \zeta \cdot 2\sqrt{mk} = 667$ [Ns/m]

m11d04B0401

(1)	$\zeta = \frac{1}{2M_a} = 0.250$
(2)	$c = \zeta \cdot 2\sqrt{mk} = 866$ [Ns/m]

フリースペース（質問などどうぞ）